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| **Fiscal Year:** | FY 2017 | **Task Last Updated:** | FY 08/01/2016 |
| **PI Name:** | Scott, Jessica Ph.D. | | |
| **Project Title:** | Influence of Exercise Modality on Cerebral-Ocular Hemodynamics and Pressures | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | (1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes<br>(2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS) | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | scottj1@mskcc.org | **Fax:** | FY |
| **PI Organization Type:** | NON-PROFIT | **Phone:** | |
| **Organization Name:** | Memorial Sloan Kettering Cancer Center | | |
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| **City:** | New York | **State:** | NY |
| **Zip Code:** | 10065 | **Congressional District:** | 12 |
| **Comments:** | NOTE (Ed., 8.1.18): Moved to Memorial Sloan Kettering Cancer Center, New York, NY, in summer 2017; formerly at Universities Space Research Association, Houston | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2013 HERO NNJ13ZSA002N-Crew Health OMNIBUS |
| **Start Date:** | 10/01/2014 | **End Date:** | 10/01/2016 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Allcorn, Aaron | **Contact Phone:** | 281.244.8402 |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Haykowsky, Mark Ph.D. ( University of Alberta )<br>Martin, David B.A. ( Wyle Laboratories, Inc. )<br>Ploutz-Snyder, Lori Ph.D. ( Universities Space Research Association )<br>Ploutz-Snyder, Robert Ph.D. ( Universities Space Research Association )<br>Stenger, Michael Ph.D. ( Wyle Laboratories, Inc. )<br>Ozgur, Omar M.D. ( New York Eye and Ear Infirmary )<br>Hamilton, Scott M.D. ( New York Eye and Ear Infirmary ) | | |
| **Grant/Contract No.:** | Internal Project | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | CENTRAL OBJECTIVES: To date, 19 out of 25 long-duration crew members (76%) have experienced in-flight and/or post-flight vision changes. These changes define the visual impairment/intracranial pressure (VIIP) syndrome. Although the exact cause of VIIP is unknown at this time, it is suspected that the microgravity-induced shift in fluids from the lower body to the upper body (cephalad fluid shift) plays a significant role. This fluid shift, in turn, may cause an elevation in intracranial pressure (ICP) and intraocular pressure (IOP). Another factor that has been proposed to contribute to VIIP is exercise. Although moderate and high intensity aerobic or resistance exercise have clearly identified benefits for cardiac, muscle, and bone health, whether such exercise contributes to the development of VIIP is unknown.<br>METHODS: Our overall goal is to characterize the impact of 3 exercise modalities used by astronauts on cerebral blood flow, ICP, and IOP. We propose to use head down tilt (HDT), a ground based analog that is well established to elicit similar cephalad fluid shifts as spaceflight. Subjects will undergo 3 HDT sessions: 1) HDT + resistance exercise, 2) HDT + moderate intensity aerobic exercise, and 3) HDT + high intensity aerobic exercise. During and following each HDT session cerebral blood flow, IOP, and ICP will be measured.<br>SIGNIFICANCE: Information characterizing factors contributing to the VIIP syndrome is of fundamental importance for sustaining human presence in space and extending the exploration of our Solar system. NASA's Human Research Program (HRP) has therefore established risks and gaps related to determining the etiology of visual acuity and ocular structural and functional changes observed in- and post-flight, and identifying safe and effective countermeasure to mitigate changes in ocular structure and intracranial hypertension. This proposal addresses the NASA request for short-term proposals that could lead to novel breakthroughs addressing one or more risks and gaps. Our proposal is specifically relevant for: Risk of Spaceflight-Induced Intracranial Hypertension/Vision Alterations and the following Gaps: Gap VIIP1: What are the etiological mechanisms and contributing risk factors for ocular structural and functional changes seen in-flight and post-flight? Gap VIIP13: Identify preventative and treatment countermeasures to mitigate changes in ocular structure and function and intracranial pressure during spaceflight. It is expected that results from the proposed investigation will provide important information that could ultimately not only improve the well being of astronauts in microgravity and upon return to Earth, but could also enhance the well-being of numerous populations such as individuals with intracranial hypertension and glaucoma. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Research Impact: This project will provide essential data and methods to quantify the impact of exercise on ICP and IOP during a cephalad fluid shift. Additionally, results from this investigation will provide important information to protect the health and mission readiness of current International Space Station (ISS) crew and to safeguard the fitness of even longer duration astronauts for Moon and Mars missions.<br>Earth Benefits: There is currently no evidence on concurrent cerebral and ocular hemodynamics and pressures during exercise in the upright or spaceflight analog conditions. This data will enable accurate assessment of exercise-induced differences in cerebral-ocular hemodynamics and pressures between exercise in a 1G environment and exercise in the spaceflight environment. It is expected that results from the proposed investigation will also provide information that could enhance the well-being of numerous clinical populations such as individuals with intracranial hypertension and glaucoma. We are also testing novel technology that could be used in clinical settings. Specifically, we evaluated acquisition of IOP continuously using a contact lens (Triggerfish, Sensimed), and the reliability and validity of a 3D imaging tool to quantify cephalad fluid shift induced facial edema. These tools could be applied to a patient being followed for thyroid eye disease, or a space occupying lesion such as an orbital tumor producing proptosis. |
| **Task Progress:** | Year 2 overview:<br>To characterize variability in internal jugular vein (IJV) flow and to create a standardized protocol for IJV acquisition, we assessed IJV flow in 10 healthy adults at -20º, 0º, and +90º. Results indicate that in order to minimize systematic error, blood velocity acquisition should encompass the vessel lumen close to omohyoid muscle.<br>Data was acquired in 10 additional subjects to:<br>1) Assess estimated intraocular pressure (IOP) continuously during exercise using Sensimed's Triggerfish contact lens system.<br>2) Determine whether raising IOP during exercise with swimming goggles rebalances translaminar pressure gradient.<br>3) Evaluate a 3-dimensional (3-D) imaging tool to provide an estimate of facial edema.<br>Analysis is ongoing. |
| **Bibliography Type:** | Description: (Last Updated: 09/05/2019) |
| **Abstracts for Journals and Proceedings** | Scott JM, Martin D, Crowell B, Goetchius E, Seponski C, Gonzales R, Matz T, Ploutz-Snyder R, Ploutz-Snyder L. "Influence of Exercise Modality on Cerebral-Ocular Hemodynamics and Pressures." Presented at 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016.<br>2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016 |